

SQL Introduction

Learning Objectives

- Define SQL and its role as a standard database language
- Identify the main categories of SQL commands
- Write and execute a basic SQL statement to create a table

Engage (5-7 minutes)

Imagine asking a librarian for all books by a specific author. You speak in a structured way the librarian understands. SQL works the same way—it is the language you use to talk to a database and request specific data.

Explore (10-12 minutes)

Examine sample SQL commands provided: CREATE TABLE, INSERT INTO, SELECT, UPDATE, DELETE. Group them into categories—Data Definition Language (DDL) and Data Manipulation Language (DML). Discuss which commands change structure and which change content.

Explain (10-12 minutes)

SQL (Structured Query Language) is the standard language for managing relational databases. DDL commands like CREATE, ALTER, and DROP define table structures. DML commands like SELECT, INSERT, UPDATE, and DELETE manipulate the data inside those tables. SQL is case-insensitive but keywords are conventionally written in uppercase.

Elaborate (8-10 minutes)

Write a CREATE TABLE statement for a “Students” table with fields: StudentID (INTEGER, primary key), FirstName (TEXT), LastName (TEXT), DateOfBirth (DATE), and Grade (TEXT). Explain each data type choice and why StudentID must be unique.

Evaluate (5-8 minutes)

Worksheet: Categorize ten given SQL commands into DDL or DML. Write a CREATE TABLE statement for a “Products” table with at least four appropriate fields.